ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba

Unit 7.01: Triangle Centers, Ceva, Menelaus & Stewart Theorem Name: _____

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 18x + 98 in the domain of Triangle Centers, Ceva, Menelaus & Stewart. By completing the square f_1x = x - h^2 + k , find the sum k + f_111 .

[ ] A: 47
[ ] B: 21
[ ] C: 38
[ ] D: 36

Q2. A mathematical model in Triangle Centers, Ceva, Menelaus & Stewart is defined by the cubic function y_2x = 2x^3 - 9x^2 + 15x - 14 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 69
[ ] B: 0
[ ] C: 15 + 1
[ ] D: 15

Q3. In Triangle Centers, Ceva, Menelaus & Stewart, two displacement vectors are given by mathbf{u}_{3} = langle 9, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 170
[ ] B: 25
[ ] C: 145
[ ] D: 282

Q4. The linear growth profile shown in the diagram for Triangle Centers, Ceva, Menelaus & Stewart follows an arithmetic progression with first term a_1 = 10 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 94
[ ] B: 780
[ ] C: 870
[ ] D: 1410

Q5. In Triangle Centers, Ceva, Menelaus & Stewart, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 36x,mathrm{d}x .

[ ] A: 144
[ ] B: 156
[ ] C: 120
[ ] D: 61

Q6. In Triangle Centers, Ceva, Menelaus & Stewart, consider the quadratic function f_6x = x^2 - 20x + 189 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_617 .

[ ] A: 237
[ ] B: 138
[ ] C: 220
[ ] D: 227

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Q7. The graph illustrates the cubic model y_7x = 2x^3 - 10x^2 + 33x - 21 studied in Triangle Centers, Ceva, Menelaus & Stewart. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 257

[ ] B: 417

[ ] C: 224

[ ] D: 61

Q8. In Triangle Centers, Ceva, Menelaus & Stewart, two displacement vectors are given by mathbf{u}_{8} = langle 10, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 45

[ ] B: 194

[ ] C: 149

[ ] D: 264